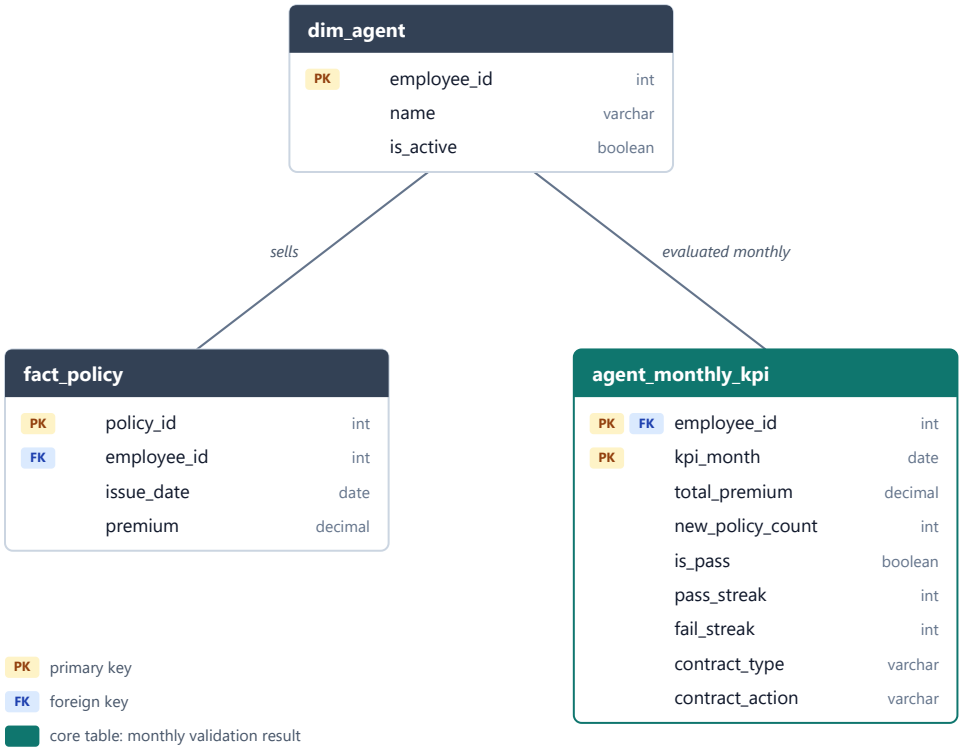
Test Case #1 — Data Modeling: Agent KPI Validation

ER (entity-relationship) diagram



Tables

- `dim_agent` : one row per agent. `is_active` decides who gets evaluated each month.
- `fact_policy` : one row per policy sold. `issue_date` decides which month it counts toward; `premium` is summed for the KPI.
- `agent_monthly_kpi` : one row per active agent per month, including zero-sales months. Totals the month's policies and premium, marks pass/fail, updates the pass/fail streaks, and records the contract held plus any switch triggered when a streak reaches 3.

Monthly process

1. At month end, create one `agent_monthly_kpi` row for every active agent.
2. Fill `total_premium` and `new_policy_count` from `fact_policy` (0 if no sales).
3. Set `is_pass` (`total_premium > 15000 AND new_policy_count > 5`), then the streaks from the previous month's row.
4. Set `contract_type` from the previous month's row (applying its `contract_action`), or `SALARY` for a new agent.
5. If `fail_streak >= 3` and the contract is `SALARY` → `TO_COMMISSION`. If `pass_streak >= 3` and the contract is `COMMISSION` → `TO_SALARY`. Otherwise `NONE`.

Assumptions

- **Strict thresholds**: exactly 15,000 baht or exactly 5 policies fails, because the brief says "more than".
- **New policy** = any policy issued in the month. **Premium** = the premium amount recorded on each policy.
- **Zero-sales month = fail.**
- **New agents start on a salary-based contract** (not specified in the brief).
- **Streaks do not reset on a contract switch.** A 4th fail while already on commission just gives `contract_action = NONE`.
- **A contract change takes effect the following month**: the next row's `contract_type` shows the new contract.

Data dictionary

dim_agent

Name	Data type	Key	Example
employee_id	int	PK	1001
name	varchar(100)		Somchai Jaidee
is_active	boolean		true

fact_policy

Name	Data type	Key	Example
policy_id	int	PK	500123
employee_id	int	FK → dim_agent	1001
issue_date	date		2026-10-14
premium	decimal(12,2)		3200.00

agent_monthly_kpi

Name	Data type	Key	Example
employee_id	int	PK, FK → dim_agent	1001
kpi_month	date	PK	2026-10-01
total_premium	decimal(12,2)		18500.00
new_policy_count	int		6
is_pass	boolean		true
pass_streak	int		3
fail_streak	int		0
contract_type	varchar(20)		COMMISSION
contract_action	varchar(20)		TO_SALARY

`contract_type` : SALARY / COMMISSION. `contract_action` : NONE / TO_COMMISSION / TO_SALARY. PK = primary key, FK = foreign key.